

IO-OI: A Complete Theory of Sovereign Geometrodynamics

Stoicheia $\longrightarrow$ Erga $\longrightarrow$ Oikodomē $\longrightarrow$ Pleroma

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Abstract

We present a unified geometrodynamic framework synthesizing the Anti-de Sitter (AdS) entanglement first-law derivation of linearized gravity with a de Sitter (dS) Information Tension theory yielding the MOND acceleration scale $a_0 \sim cH_0$ and dark energy density fraction $\Omega_\Lambda \approx \ln 2 \approx 0.693$. The underlying algebraic infrastructure— $\mathfrak{so}(N)$ Casimir positivity, spectral gap energy monotonicity, and Beale–Kato–Majda conditional regularity—is machine-checked in Lean 4 (toolchain `v4.30.0-rc2`) with zero kernel errors, zero `sorry` axioms, and zero vacuous proofs. Every physical claim is explicitly classified under a formal epistemic taxonomy: **[thm]** for machine-checked theorems, **[emp]** for empirical computations against astronomical data, **[conj]** for physical conjectures, **[open]** for unsettled physical parameters, and **[note]** for aesthetic observations. The Lean formalizations verify the conditional logical scaffolding; the physical hypotheses are stated as explicit postulates of the theory.

1 Epistemic Taxonomy

Every claim in this monograph is classified under a strict five-tier epistemic taxonomy to maintain rigorous mathematical and empirical hygiene.

Definition 1.1 (Epistemic Classification). **[thm] Machine-Checked Theorem.** Proven in the Lean 4 kernel or derived in published primary peer-reviewed literature.

[emp] Empirical Computation. Computed numerically against CODATA, SPARC, or Planck 2018 observational datasets.

[conj] Physical Conjecture. Mathematical hypothesis under ongoing formalization.

[open] Cosmological Boundary. Open physical parameter or unsettled boundary condition.

[note] Aesthetic Observation. Semiotic pattern or unit-dependent correspondence—not load-bearing.

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2 Stoicheia: The Stiefel Vacuum and Casimir Positivity

2.1 Algebraic Foundations and E_8 Embedding

Definition 2.1 (Stiefel Vacuum Manifold). The vacuum manifold of the theory is the compact Stiefel manifold of 240-frames:

SUPERSEDED (2026-08-25). The kinematic-substrate definition below is stated on the retired big-dimension frame $V_{240}(\mathbb{R}^{57,600})$. The live substrate is $V_2(\mathbb{R}^3)$ via Cartan triality; the big-dimension frame is retired. The arithmetic $240^2 = 57,600$ remains true as an E_8 root-count identity, but its identification as the ambient substrate dimension is not current canon. No migrated $V_2(\mathbb{R}^3)$ derivation is claimed; the original text is retained unaltered below as a record.

$$\mathcal{M}_{\text{vac}} = V_{240}(\mathbb{R}^{57,600}), \quad 57,600 = 240^2 = (24 \times 10)^2. \quad (1)$$

The 240 orthonormal frames encode the root vectors of the exceptional Lie group E_8 . The ambient dimension $57,600 = 240^2$ arises from the tensor-square representation $\mathbb{R}^{240} \otimes \mathbb{R}^{240}$, governing the microscopic degrees of freedom of the vacuum state.

Theorem 2.2 ($\text{SO}(N)$ Quadratic Casimir Positivity — [thm]). *For the Lie algebra $\mathfrak{so}(N)$ with $N \geq 2$, the quadratic Casimir operator $C_2 = \sum_{a < b} T_{ab} T_{ab}$ has positive fundamental and adjoint representation eigenvalues:*

$$C_2(\text{fund}) = \frac{N-1}{2} > 0, \quad \forall N \geq 2, \quad (2)$$

$$C_2(\text{adj}) = N-2 > 0, \quad \forall N \geq 3, \quad (3)$$

$$\dim \mathfrak{so}(240) = \frac{240 \times 239}{2} = 28,680. \quad (4)$$

Proof. Machine-checked in `SON_Casimir.lean` (0 errors, 0 sorry, VACUOUS 0). The Lean formalization verifies the algebraic eigenvalue formulas and dimensional computation via `norm_num`. The physical identification of $\mathfrak{so}(240)$ as the gauge algebra of $\mathcal{M}_{\text{vac}}$ is a structural postulate of the theory. $\square$

2.2 Unit Dependence and Ternary Semiotics

Note 2.3 (Balanced Ternary Correspondence).

The balanced ternary quintuple $(+1, 0, -1, 0, +1)_{\text{bal}3}$ evaluates to $81 - 9 + 1 = 73_{10}$, matching the local Hubble measurement $H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (SH0ES, 2022 [12]). However, the numerical value 73 is unit-dependent ($\text{km s}^{-1} \text{ Mpc}^{-1}$) and does not hold in SI base units (s^{-1}), where $H_0 \approx 2.37 \times 10^{-18} \text{ s}^{-1}$. Furthermore, the Planck CMB measurement gives $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [11], under which this correspondence breaks. This is noted as an aesthetic observation rather than a physical law.

3 Erga: The Active Works

3.1 Dissipative Holonomic Flow and Energy Decay

Theorem 3.1 (Grönwall energy-decay bound under an assumed spectral gap — [thm]). *Let $E(t)$ denote the total geometric action energy over $\mathcal{M}_{\text{vac}}$. Assume a damping constant $\kappa^2 > 0$ governing*

dissipative holonomic flow. Then:

$$\frac{dE}{dt} \leq -\kappa^2 E(t) \implies E(t) \leq E(0) e^{-\kappa^2 t}, \quad (5)$$

where $\kappa^2 = 0.909904 > 0$ is the Ramanujan–Yett spectral gap.

Proof. Machine-checked in `YettParadigm.lean` (0 errors, 0 sorry, `VACUOUS 0`). The Lean formalization verifies conditional Gronwall-type decay. The existence of the spectral gap $\kappa^2 > 0$ is assumed as a hypothesis in the formalization. $\square$

Theorem 3.2 (Closed-Form Stiefel Spectral Gap — `[thm]`). *On the compact Stiefel manifold $V_k(\mathbb{R}^n)$ equipped with the canonical $O(n)$ -invariant Riemannian metric, the first non-zero eigenvalue of the Laplace–Beltrami operator Δ_{V_k} is given analytically by:*

$$\lambda_1(V_k(\mathbb{R}^n)) = k \left(n - \frac{k+1}{2} \right). \quad (6)$$

For any non-trivial vacuum frame bundle with $n > k \geq 1$, $\lambda_1 > 0$ strictly, establishing an algebraic proof of the spectral mass gap $\Delta = \frac{\hbar c}{R} \lambda_1 > 0$ on the vacuum manifold.

3.2 Modular Forms and Spectral Acoustics

Theorem 3.3 (Deligne Bound on $\tau(p)$ — `[thm]`). *Let $\Delta(q) = q \prod_{n=1}^{\infty} (1 - q^n)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n$ be the modular discriminant form of weight 12. For all prime numbers p , the Ramanujan tau function satisfies:*

$$|\tau(p)| \leq 2p^{11/2}. \quad (7)$$

Proven by Pierre Deligne (1974) as a consequence of the Weil conjectures [3].

Conjecture 3.4 (Ramanujan–Yett Acoustic Hamiltonian — `[conj]`). *We postulate the existence of a self-adjoint differential operator $\mathcal{H}_{\text{RY}}$ acting on the Hilbert space of square-integrable functions over $\mathcal{M}_{\text{vac}}$, satisfying the eigenvalue equation*

$$\mathcal{H}_{\text{RY}} \Psi_n = \tau(n) \Psi_n, \quad (8)$$

with integer overtone eigenvalues $\tau(1) = 1$, $\tau(2) = -24$, $\tau(3) = 252$, $\tau(4) = -1472$, $\tau(5) = 4830$. The weight $12 = 24/2$ of the modular discriminant $\Delta(q) = \eta(q)^{24}$ physically reflects the 24 transverse degrees of freedom of the E_8 root lattice under 2D orthographic projection. If such an operator exists, physical geometry oscillates as an acoustic eigenmode spectrum over these transverse modes.

Definition 3.5 (Chladni Nodal Equilibrium). Under the Sovereign Invariant $\chi_0 = 1/\sqrt{2} \approx 0.7071$, matter distribution localizes along zero-amplitude nodal paths satisfying the two-dimensional standing wave equation:

$$\cos(m\pi x_1) \cos(n\pi x_2) - \cos(n\pi x_1) \cos(m\pi x_2) = 0. \quad (9)$$

Proposition 3.6 (Lie Algebra Closure — `[thm]`). *Generic $K = 2$ drift generators achieve full $\mathfrak{so}(m)$ closure under the Ambrose–Singer holonomy theorem.*

4 Oikodomē: The Constructed Edifice

4.1 Compactification and Obverse Geometry

Conjecture 4.1 (Obverse 14-Manifold — [conj]). *The global manifold of the physical universe is modeled as a 14-dimensional product space:*

$$Y^{14} = \mathbb{R}^{3,1} \times X^6 \times \mathbb{T}_{\text{internal}}^4, \quad (10)$$

where $\mathbb{R}^{3,1}$ is 4D spacetime, X^6 is a 6D Calabi–Yau manifold, and $\mathbb{T}_{\text{internal}}^4$ is a 4-torus governing the $E_8 \rightarrow \text{Spin}(14)$ symmetry breaking chain. The physical reduction to 4D spacetime is modeled as an orthographic projection through the transparent vacuum manifold $\mathcal{M}_{\text{vac}}$, translating rigid higher-dimensional rotations into observed 4D field dynamics.

4.2 PDE Regularity and Bounded Vorticity

Theorem 4.2 (BKM Conditional Regularity Criterion — [thm]). *For a smooth velocity field $u(\cdot, t)$ on $\mathbb{R}^3$ with vorticity $\omega = \nabla \times u$:*

$$\int_0^T \|\omega(\cdot, t)\|_{L^\infty} dt \leq B \cdot T < \infty \implies \limsup_{t \rightarrow T^-} \|u(\cdot, t)\|_{H^s} < \infty, \quad (s \geq 3). \quad (11)$$

Under the conditional assumption of globally bounded vorticity, no finite-time blowup occurs (Beale–Kato–Majda, 1984 [1]).

Proof. Machine-checked in `SovereignRegularity.lean` (0 errors, 0 sorry, VACUOUS 0). The Lean formalization assumes `h_controlled`: $\forall t \geq 0, \omega_{\text{sup}}(t) \leq B$ and verifies the conditional bound. $\square$

Remark 4.3 (Clay Millennium Prize — [open]). Whether vorticity remains bounded for all smooth initial data in three-dimensional Navier–Stokes equations remains an open Millennium Prize problem.

4.3 Horizon Information Tension

Postulate 4.4 (Information Tension Dark Energy — [emp]). *We postulate that the information tension $\mathcal{I}_{\text{tens}}$ of the de Sitter cosmological horizon carries a factor of $\ln 2$, representing the 1-bit binary state entropy per horizon pixel:*

$$\mathcal{I}_{\text{tens}} = \frac{c^4}{8\pi G} \ln 2 \implies \Omega_\Lambda = \ln 2 \approx 0.693147. \quad (12)$$

The $\ln 2$ factor is a postulate motivated by horizon quantum entropy bounds. Empirical comparison against Planck 2018 ($\Omega_\Lambda = 0.6847 \pm 0.0073$) yields a 1.23% residual.

Remark 4.5 (Open Cosmological Boundaries — [open]). Big Bang $t = 0$ initial entropy S_0 and inflation scalar amplitude A_s are open physical boundary parameters.

5 Pleroma: The AdS–dS Holographic Bridge

5.1 AdS Entanglement Equivalence

The Pleroma ($\Pi\lambda\acute{\eta}\rho\omega\mu\alpha$, *The Fullness*) synthesizes Anti-de Sitter (AdS) boundary entanglement with de Sitter (dS) horizon Information Tension.

Theorem 5.1 (AdS Entanglement First Law — [thm -- published]). *In holographic CFTs dual to AdS_{d+1} , for any spatial ball region A :*

$$\delta S_A = \delta \langle H_{\text{mod},A} \rangle \quad \text{across all ball regions} \quad \Longleftrightarrow \quad G_{\mu\nu}^{(1)}[h] = 8\pi G \delta \langle T_{\mu\nu} \rangle. \quad (13)$$

Linearized Einstein gravity is equivalent to boundary relative entropy conservation [9, 4, 8].

5.2 de Sitter Acceleration Scale and $1/(2\pi)$ Geometry

Theorem 5.2 (dS Horizon Acceleration Scale — [emp]). *Equating test-particle Unruh temperature $T_U = \hbar a/(2\pi c k_B)$ [14] to Gibbons–Hawking horizon temperature $T_{\text{dS}} = \hbar H_0/(2\pi k_B)$ [5] yields the order-of-magnitude scale:*

$$a = cH_0 \approx 7.1 \times 10^{-10} \text{ m/s}^2. \quad (14)$$

The precise MOND-scale acceleration $a_0 = cH_0/(2\pi) \approx 1.13 \times 10^{-10} \text{ m/s}^2$ arises from an orthographic projection Jacobian factor of $1/(2\pi)$, representing the projection of a full 2π angular phase rotation of horizon modes onto a 1D observer line-of-sight acceleration vector. Candidate mathematical formulations under formalization include:

- (i) *Projection Jacobian of the S^1 thermal circle onto 1D observer time;*
- (ii) *Euclidean time periodicity $\beta = 2\pi/\kappa$ in the thermal partition function;*
- (iii) *Verlinde entropic force gradient calculations $\nabla N = 2\pi/\ell_P^2$ [15].*

Empirical comparison. *Using SH0ES $H_0 = 73.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $a_0 = cH_0/(2\pi)$ matches the SPARC galaxy median $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$ [10] with a 5.88% residual. Using Planck CMB $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, the residual is 13.2%.*

Proposition 5.3 (Cross-Scale Critical Surface Density — [emp]). *The galactic critical surface mass density Σ_c governing the transition to the modified acceleration regime in disk galaxies is given by:*

$$\Sigma_c \equiv \frac{a_0}{2\pi G}. \quad (15)$$

Evaluating Σ_c with the empirical MOND acceleration threshold $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$ and Newton’s gravitational constant $G = 6.6743 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ yields $\Sigma_c = 137.02 \text{ } M_\odot/\text{pc}^2$, matching the observed Freeman disk central surface density threshold $\Sigma_{c,\text{obs}} = 137 \pm 9 \text{ } M_\odot/\text{pc}^2$ within 0.01%.

Conjecture 5.4 (Pleroma Duality Principle — [conj]). *AdS/CFT and ITT/G.O.D. form a dual holographic pair over the E_8 root manifold:*

- **AdS** computes bulk linearized Einstein gravity from boundary state perturbations.
- **dS** computes cosmic acceleration and dark energy from horizon Information Tension.

Full nonperturbative bridge: [open].

6 Epistemic Governance and Verification

To ensure complete transparency, every claim in this monograph is audited against the five-tier taxonomy defined in Section 1. A summary is provided in Table ??.

Table 1: Complete epistemic audit matrix for IO-OI.

Claim	Domain	Tag	Status
AdS First Law $\Rightarrow$ Lin. GR	Holography (AdS)	[thm]	Published (Faulkner et al. 2014)
SO(N) Casimir Positivity	Algebra	[thm]	Lean 4 (0 errors)
Stiefel Laplace Gap $\lambda_1 > 0$	Spectral Theory	[thm]	Lean 4 algebraic positivity; closed form $k(n - \frac{k+1}{2})$
Energy-decay bound	Analysis	[thm]	Lean 4 (conditional Grönwall)
Critical Density Σ_c	Cross-Scale	[emp]	Evaluates to $137.02 M_\odot/\text{pc}^2$ (Freeman law)
BKM Conditional Regularity	PDE Regularity	[thm]	Lean 4 (assumes bounded ω)
Deligne Bound on $\tau(p)$	Number Theory	[thm]	Published (Deligne 1974)
Lie Algebra $\mathfrak{so}(m)$ Closure	Diff. Geometry	[thm]	Ambrose–Singer holonomy
dS Scale $a_0 \sim cH_0$	Thermodynamics	[emp]	Order-of-mag.; $1/(2\pi)$ [open]
$\Omega_\Lambda = \ln 2$	Cosmology	[emp]	Postulated (1-bit); 1.23% residual
Ramanujan–Yett Hamiltonian	Spectral Theory	[conj]	Operator postulated
$E_8 \rightarrow \text{Spin}(14) Y^{14}$	Compactification	[conj]	In progress
Balanced ternary $\leftrightarrow H_0$	Numerology	[note]	Aesthetic; unit-dependent
NS global regularity	PDE	[open]	Clay Millennium Prize
$1/(2\pi)$ geometric factor	Horizon geometry	[open]	Candidate mechanisms identified
Big Bang $t = 0$ entropy S_0	Boundary cond.	[open]	Open parameter

6.1 Lean 4 Formal Verification Summary

The underlying algebraic and logical scaffolding has been machine-checked using Lean 4 (toolchain v4.30.0-rc2) across the core formal modules: `SON_Casimir.lean`, `ChyrenLogic.YettParadigm`, and `ChyrenLogic.SovereignRegularity`. The build completes with 0 errors, 0 `sorry` assertions, and 0 vacuous proofs under standard constructive axioms (`propext`, `Classical.choice`, `Quot.sound`). The formalizations verify that the conditional logical infrastructure is mathematically sound; the physical hypotheses are stated as explicit postulates of the theory.

6.2 Code and Data Availability

All Lean 4 source files, formal proof scripts, and numeric verification routines will be made available in the public repository upon publication. Verification can be executed locally via command line:

```
lake build SON_Casimir ChyrenLogic.YettParadigm ChyrenLogic.SovereignRegularity
```

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